WORKED SOLUTION

STEP II 1991, Question 2

Pure • Asymptotes of a Hyperbola

STEP II 1991, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q2

The key idea

The quadratic part $3y^2 - 10xy + 3x^2$ factorises as $(3y - x)(y - 3x)$, and this is the whole story: once you complete the factorisation to include the linear terms, the curve becomes "product of two linear expressions $= 1$". Far out along the curve one factor must blow up, so the other must tend to zero, and that factor, set equal to zero, is an asymptote. This is exactly the "relationship between s and t in the limit" that the hint asks for.

Solution

The tangent

Differentiating $(\dagger)$ implicitly with respect to x,

$$6y\frac{\mathrm{d}y}{\mathrm{d}x} - 10y - 10x\frac{\mathrm{d}y}{\mathrm{d}x} + 6x + 16\frac{\mathrm{d}y}{\mathrm{d}x} - 16 = 0,$$

so

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{10y - 6x + 16}{6y - 10x + 16} = \frac{5y - 3x + 8}{3y - 5x + 8}.$$

At (s, t) the gradient is $m = \dfrac{5t - 3s + 8}{3t - 5s + 8}$, and the tangent is $y = mx + (t - ms)$. Putting the intercept over the common denominator,

$$t - ms = \frac{t(3t - 5s + 8) - s(5t - 3s + 8)}{3t - 5s + 8} = \frac{3t^2 - 10st + 3s^2 + 8t - 8s}{3t - 5s + 8}.$$

Since (s, t) lies on the curve, $3t^2 - 10st + 3s^2 = -16t + 16s - 15$, so the numerator is $-8t + 8s - 15 = -(8t - 8s + 15)$. Hence the tangent is

$$y = \left(\frac{5t - 3s + 8}{3t - 5s + 8}\right)x - \left(\frac{8t - 8s + 15}{3t - 5s + 8}\right),$$

as required. (Where $3t - 5s + 8 = 0$ the tangent is vertical, and the formula is to be read in the cleared-denominator form.)

The asymptotes

Try to write (†) as $(3y - x + a)(y - 3x + b) + k = 0$. Expanding,

$$(3y - x + a)(y - 3x + b) = 3y^2 - 10xy + 3x^2 + (3b + a)y - (b + 3a)x + ab,$$

so we need $3b + a = 16$ and $b + 3a = 16$, giving $a = b = 4$ and $ab = 16$. Therefore (†) is

$$(3y - x + 4)(y - 3x + 4) = 1.$$

For the point (s, t) write

$$u = 3t - s + 4, \qquad v = t - 3s + 4, \qquad \text{so that} \qquad uv = 1.$$

Every expression in the tangent is a combination of u, v and constants. Matching coefficients of t and s gives

$$\begin{aligned} 5t - 3s + 8 &= \tfrac{1}{2}(3u + v), \\ 3t - 5s + 8 &= \tfrac{1}{2}(u + 3v), \\ 8t - 8s + 15 &= 2(u + v) - 1 = 2(u + v) - uv, \end{aligned}$$

using $uv = 1$ in the last one. So the gradient and intercept of the tangent are

$$m = \frac{3u + v}{u + 3v}, \qquad c = \frac{8t - 8s + 15}{3t - 5s + 8} = \frac{2(2u + 2v - uv)}{u + 3v}.$$

Ask yourself As $s \to \infty$, what happens to u and v?

Eliminating t, $u - 3v = 8s - 8$, so $u - 3v \to \infty$. Since $uv = 1 > 0$, u and v have the same sign.

- If both are positive, $u - 3v \to \infty$ forces $u \to \infty$, and then $v = 1/u \to 0$.
- If both are negative, $u - 3v = 3|v| - |u| \to \infty$ forces $|v| \to \infty$, and then $u = 1/v \to 0$.

This is the relationship the hint wants: on one branch $t - 3s + 4 \to 0$, on the other $3t - s + 4 \to 0$.

Case $v \to 0$, $u \to \infty$. Dividing top and bottom by u,

$$m = \frac{3 + v/u}{1 + 3v/u} \to 3, \qquad c = \frac{2(2 + 2v/u - v)}{1 + 3v/u} \to 4,$$

so the limiting tangent is $y = 3x - 4$.

Case $u \to 0$, $v \to \infty$. Dividing top and bottom by v,

$$m = \frac{3u/v + 1}{u/v + 3} \to \tfrac{1}{3}, \qquad c = \frac{2(2u/v + 2 - u)}{u/v + 3} \to \tfrac{4}{3},$$

so the limiting tangent is $y = \frac{1}{3}x - \frac{4}{3}$, that is, $3y = x - 4$.

So the asymptotes are $y = 3x - 4$ and $3y = x - 4$. (Letting $s \to -\infty$ instead gives the same two lines with the roles of the cases swapped.) Notice that these are exactly the lines $v = 0$ and $u = 0$: the factorised form shows them straight away.

Angles and axes

The line $y = 3x - 4$ has gradient 3, so it makes an angle θ with the x-axis where $\tan\theta = 3$. Its angle with the y-axis is then $\frac{\pi}{2} - \theta$, and $\tan(\frac{\pi}{2} - \theta) = \cot\theta = \frac{1}{3}$. The line $3y = x - 4$ has gradient $\frac{1}{3}$, so its angle with the x-axis is $\arctan\frac{1}{3}$, the same as the angle between the first asymptote and the y-axis.

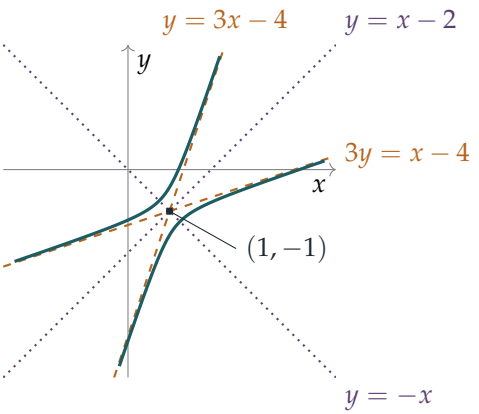
? **Ask yourself** Why does this tell us the bisectors?

Reflection in the line $y = x$ swaps the x- and y-axes, and by what we've just shown it swaps the directions of the two asymptotes: direction $(1,3)$ goes to $(3,1)$. The angle between the asymptotes is therefore symmetric about the direction of $y = x$, which bisects it: the asymptote directions are at $\arctan 3$ and $\arctan\frac{1}{3}$ from the x-axis, and these are $\frac{\pi}{4} \pm (\arctan 3 - \frac{\pi}{4})$. The other bisector is perpendicular to the first. So the bisectors have gradients 1 and -1.

The axes of a hyperbola are the angle bisectors of its asymptotes through its centre, the point where the asymptotes meet. Solving $y = 3x - 4$ and $3y = x - 4$: $9x - 12 = x - 4$, so $x = 1$, $y = -1$. The centre is $(1, -1)$, and the axes are

$$y + 1 = x - 1 \iff y = x - 2, \qquad y + 1 = -(x - 1) \iff y = -x.$$

To see which is which: on $y = -x$ we get $u = v = 4 - 4x$, so $uv = 1$ gives $x = 1 \pm \frac{1}{4}$, and the vertices are $(\frac{5}{4}, -\frac{5}{4})$ and $(\frac{3}{4}, -\frac{3}{4})$. So $y = -x$ is the transverse axis. On $y = x - 2$ we get $u = 2x - 2 = -v$, so $uv = -(2x - 2)^2 \leq 0$ and that line (the conjugate axis) never meets the curve.



Checks

- The point $(\frac{5}{4}, -\frac{5}{4})$ satisfies (†): $3 \cdot \frac{25}{16} + 10 \cdot \frac{25}{16} + 3 \cdot \frac{25}{16} - 20 - 20 + 15 = 25 - 25 = 0$.
- Numerically, taking $u = 10^8$ (so $s \approx 1.25 \times 10^7$), the tangent formula gives gradient 3.0000000 and intercept term 3.99999998; taking $u = -10^{-8}$ gives 0.33333333 and 1.33333334.

- The centre $(1,-1)$ gives $u = v = 0$, so it is not on the curve, which is why the tangent formula never has $0/0$.

Common slips

- Finding only the gradients 3 and $\frac{1}{3}$ from $3k^2 - 10k + 3 = 0$ (with $t \approx ks$) and forgetting that the intercept needs the next term: $t = 3s - 4 + o(1)$, not just $t \approx 3s$.
- Taking the bisectors through the origin rather than through the centre $(1,-1)$.

Beyond STEP

Every conic $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ can be written as $\mathbf{x}^{\mathsf{T}} A\mathbf{x} + \ldots$ with a symmetric matrix $A = \begin{pmatrix} a & h \\ h & b \end{pmatrix}$, and its axes point along the eigenvectors of A. Here $A = \begin{pmatrix} 3 & -5 \\ -5 & 3 \end{pmatrix}$ has eigenvectors $(1,1)$ and $(1,-1)$, with eigenvalues -2 and 8 of opposite sign, which is what makes the curve a hyperbola. Diagonalising quadratic forms like this is a central topic of first-year linear algebra; see Strang, *Introduction to Linear Algebra*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.